

The Grotthuss mechanism

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Abstract

Suggested mechanisms for proton mobility are confronted with experimental findings and quantum mechanical calculations, indicating that no model is consistent with the existing data. It is suggested that the molecular mechanism behind prototropic mobility involves a periodic series of isomerizations between H_9O_4^+ and H_5O_2^+ , the first triggered by hydrogen-bond cleavage of a second-shell water molecule and the second by the reverse, hydrogen-bond formation process.

1. Introduction

Proton mobility in water is abnormally high [1]. At room temperature its limiting ionic conductance is about seven times that of a sodium cation, or approximately five times that of K^+ . The usual explanation is in terms of a sequence of proton-transfer reactions (proton hops) between water molecules ('prototropic mobility', 'Grotthuss mechanism' [2]). In spite of many years of hard research and heated discussions, the exact details of this fundamental process remain obscure [3–12].

The models suggested for prototropic conduction no doubt contain the right ingredients for its explanation. Unfortunately, none has the full mechanism correctly assembled. In analyzing these models, it will become obvious that they all contradict known experimental facts (some, it seems, were suggested *in spite* of experimental reality). By carefully considering a wide variety of literature data, improbable mechanisms are eliminated and a likely microscopic picture of the Grotthuss mechanism emerges. This discussion is timely, because with the rapid advance in quantum chemistry calculations of proton transfer in water

[13–18], and of Monte Carlo and molecular dynamics simulations [8,17,19], it may soon become possible to perform a sufficiently large calculation to check the proposed mechanism [18]. Reciprocally, insight from the following discussion may provide important guidelines for choosing relevant reaction coordinates in *ab initio* calculations.

2. Mechanisms of prototropic mobility

Two opposing limits of solvent involvement in prototropic mobility may be considered. Bernal and Fowler [4] pictured a proton hopping from a H_3O^+ moiety to a freely rotating nearest-neighbor water molecule. This 'complete disorder' limit presents one example for an incoherent transport mechanism. Conway and co-workers [5] suggested a 'field-induced' water rotation: in the $\text{H}_2\text{OH}^+ \dots \text{HOH}$ complex, the water is forced to rotate its lone-pair towards the hydronium.

The opposite extreme is that of a large, static water cluster within which an ordered hydrogen-bonded network exists. In a familiar picture ('the relay mech-

anism'), the proton hops in a concerted fashion across a whole array of water molecules



The rate-limiting step could then be either the proton motion itself or a 'structural diffusion' process of adding water molecules to the periphery of the cluster [6]. Eigen claimed that proton motion is rate determining in ice, explaining their observation that its mobility is one or two orders of magnitude faster than in water [6]. Consequently, 'structural diffusion' is rate limiting in water [6].

Such pictures of proton mobility dominate physical chemistry textbooks. For example, Fig. 26.4 in Atkins shows a fast relay mechanism coupled to a rate-limiting 'structural diffusion' step involving a Conway-like forced water rotation [20]. More recent discussions of proton mobility suggest proton delocalization across large water clusters [10,11]. There is ample evidence suggesting that all the above mechanisms are wrong. Such evidence is surveyed below.

2.1. Free rotors and water structure

Water is not a free rotor, but rather tightly involved in hydrogen bonding. The local structure around any given water molecule tends to be tetrahedral. There are roughly four water molecules in the first solvation shell, two donating and two accepting hydrogen bonds from the central water. X-ray diffraction yields a clear description of the oxygen–oxygen radial distribution function [21]. Its peaks correspond to nearest-neighbors and next-nearest-neighbors in tetrahedral symmetry. These peaks diminish in amplitude with increasing temperature, indicating a decreasing hydrogen-bonding content. There is evidence for the persistence of even larger ice-structure motifs in liquid water [22]. Likewise, Raman experiments [23] and molecular-dynamics simulations [19] show that in liquid water most molecules are involved in either three or four hydrogen bonds, with the enhancement of tetrahedral symmetry at lower temperatures.

On the other hand, water molecule rotation does correlate with proton mobility. Current interpretations of Raman-induced Kerr effect [23], Rayleigh light scattering [24] and inelastic neutron scattering [25] suggest that water molecule reorientation takes 1–2 ps at

room temperature. Proton hopping times as obtained by NMR and ^{17}O resonance [26] are of similar magnitude, $\tau_p \approx 1.5$ ps. These times can explain the abnormal proton mobility, see below. Thus while water molecule reorientation may be an important ingredient in proton mobility, the mechanism is not likely to involve free water rotors as suggested by Bernal and Fowler [4].

2.2. The relay mechanism and incoherent proton mobility

The persistence of structural motifs in liquid water does not mean that the relay mechanism is correct and that protons hop in a concerted fashion across large water clusters, Eq. (1). Such a mechanism contradicts so many experimental data that one wonders how it survived that long.

2.2.1. Proton mobility is incoherent

The proton does not delocalize over large water clusters.

(i) The relay mechanism predicts that proton mobility increases with increasing size of hydrogen-bonded clusters. Thus mobility should be highest in ice [6], then in water at low temperatures and pressures. Although the issue of proton conductivity in ice is still unsettled, it appears now to be about a factor of two *slower* than in super-cooled water of equal temperature [27,28]. The evidence for proton mobility in water is unequivocal, it *increases* with increasing temperature and pressure [29]. That the fraction of hydrogen-bonded water decreases with increasing temperature is clear intuitively as well as from simulations [19] and experiment [23]. Near atmospheric pressures the hydrogen-bond content decreases with increasing pressure [23], apparently because of transitions from an open water structure to a closed form with fewer interlayer hydrogen bonds [22]. Thus proton mobility increases as the hydrogen bonds weaken.

(ii) The NMR proton hopping times, τ_p , account for the abnormal proton mobility if one assumes that hopping is across a *single* water molecule at a time. Using the Einstein relation for mobility in three dimensions

$$D = l^2/6\tau_p. \quad (2)$$

Meiboom was able to estimate a reasonable proton diffusion coefficient [26]. Let us slightly modify this estimate by taking the hopping length as $l = 2.5 \text{ \AA}$, the hydrogen-bond length between water and H_3O^+ [8], rather than the water–water distance of $2.8 \text{ \AA}$. Using $\tau_p = 1.5 \text{ ps}$ gives $D = 7 \times 10^{-5} \text{ cm}^2/\text{s}$, a very reasonable estimate for the abnormal proton mobility at room temperature (subtract from the proton diffusion coefficient, $9.3 \times 10^{-5} \text{ cm}^2/\text{s}$, the water self-diffusion coefficient, $2.3 \times 10^{-5} \text{ cm}^2/\text{s}$ [1]). Even the most modest coherent effect, with proton hopping across just two water molecules, already leads to a factor of 4 in the predicted mobility. Thus proton mobility is best described as an incoherent, Markovian hopping process.

(iii) Additional evidence for the Markovian nature of the solvated proton comes from excited-state proton transfer to solvent [30]. These transient fluorescence measurements, having time resolution of about 50 ps, can be explained quantitatively by the time-dependent Smoluchowski equation describing proton diffusion in a field of force. The diffusive behavior persists under a wide variety of conditions: solvent composition, salt concentrations and pH values [30]. On even shorter timescales, femtosecond geminate recombination of proton and solvated electron in pure liquid water is interpreted [31] by diffusive proton motion on timescales of just a few ps.

2.2.2. Proton motion is not rate limiting

The possibility that proton motion within clusters is rate-limiting may be ruled-out by several considerations:

(i) The deuterium isotope effect on prototropic mobility is rather small, 1.4 at room temperature [1]. Conway and co-workers [5] estimated that proton tunneling should lead to a minimal isotope effect of 2. Westheimer and Melander have suggested that kinetic isotope effects (KIEs) are *maximal* for the symmetric, $\Delta G = 0$, reaction [32]. Proton hopping between two water molecules is expected to correspond to this symmetric limit. In this case, when a light particle is transferred between two heavy particles, the KIE is particularly large. As seen in Fig. 7 of Ref. [33], the maximal KIE for reaction series involving proton transfer is 6–10. The small isotope effect on proton mobility suggests that the reaction coordinate does not

involve proton motion.

(ii) While the activation energy of proton mobility is 2–3 kcal/mol [1], proton motion within symmetric entities such as H_5O_2^+ may be activationless. Quantum mechanical calculations show that the potential along the proton coordinate in $\text{H}_2\text{O} \dots \text{H}^+ \dots \text{OH}_2$ has a barrier only if the O–O distance exceeds its equilibrium distance of $2.39 \text{ \AA}$ [16]. At the equilibrium distance, there is no barrier for proton transfer at all [13]. The proton resides in a flat minimum inbetween the oxygens.

(iii) Intramolecular proton transfer, which *may* be limited by proton tunneling, can be much faster than the 1–2 ps timescale for proton hopping in water. Consider just one model system, 3-hydroxyflavone [34]. Direct proton transfer, observed in non-hydrogen-bonded solvents, or proton hopping across a single solvent bridge (e.g., in methanol) may be faster than 100 fs. Interestingly, in hydrogen-bonded solvents (under conditions that the donor–acceptor are unbridged) intramolecular proton transfer in 3-hydroxyflavone slows down considerably [34]. When solvent conformation is right for proton transfer, its rate can be an order of magnitude *faster* than observed in mobility measurements.

2.2.3. Cluster-growth is not rate-limiting

To the contrary, cluster degradation may be the rate-determining step: Eigen has suggested a relay mechanism with ‘structural diffusion’ as the rate-limiting step [6]. However, the hydrogen-bond content of water is so high, every molecule participating in 3–4 hydrogen bonds [19,23], that the whole body of water may be considered as one huge cluster. Since hydrogen-bond connectivity in liquid water is well above the percolation threshold, it is not clear why the formation of new hydrogen bonds at the periphery of water clusters should be rate-limiting.

2.2.4. The rate-limiting step is hydrogen-bond cleavage

Proton mobility correlates with hydrogen-bond cleavage rather than with its formation, as suggested by the fact that it *increases* with decreasing hydrogen-bond content of water, i.e. at higher temperatures and pressures [29]. The NMR hopping times [26] agree with single molecule reorientation times [23–25], a

process requiring hydrogen-bond cleavage.

The temperature dependence of the NMR [10] and proton mobility [1] data is similar, with an activation energy of 2–3 kcal/mol. This agrees with the in situ hydrogen-bond strength of 2.6 kcal/mol obtained from Raman experiments [23]. It represents the *directional* component of the hydrogen-bond enthalpy, associated with water rotation. An additional 2.6 kcal/mol is associated with isotropic electrostatic interactions, leading to a total enthalpy of about 5.2 kcal/mol per hydrogen bond, in agreement with the 10.5 kcal/mol enthalpy of vaporization of water.

The 1.4 D/H isotope effect on proton mobility [1] can also be interpreted as arising from water reorientation that leads to hydrogen-bond cleavage. Rotation around the heavy oxygen is a nearly pure hydrogen motion, so that the isotope effect should be given by the mass ratio, $\sqrt{m_D/m_H} = \sqrt{2}$. The OD...O versus OH...O enthalpy difference is thus expected to be $\Delta\Delta H = \frac{1}{2}RT \ln 2 = 0.2$ kcal/mol at room temperature. This prediction has recently been verified experimentally [23]: the temperature-dependence of the Raman OD and OH stretch bands gives $\Delta H = 2.8$ kcal/mol and 2.6 kcal/mol for deuterium- and hydrogen-bond enthalpies, respectively. While conservative error estimates on ΔH are themselves ± 0.2 kcal/mol, the difference reflected in $\Delta\Delta H$ is real.

2.3. Proton solvation and proton mobility

All models are incompatible with elementary considerations of proton solvation. Unlike water which is predominantly tetrahedral the solvated H_3O^+ cation is a nearly plane-trigonal complex (unsolvated, gas-phase H_3O^+ is pyramidal [8]). Unlike water, which is involved in 4 hydrogen bonds as both a donor and an acceptor, the positive H_3O^+ disallows hydrogen-bonding to its oxygen. This cation is best described as donating three hydrogen bonds to 3 water molecules in its first solvation shell. This forms the 'Eigen cation' [6], H_9O_4^+ , as depicted in Fig. 1.

How does the extra stability of this cation arise [13]? Gas-phase mass-spectrometric data [35] show that water ligands in the first solvation-shell are more tightly bound than those in the second and subsequent shells. The enthalpies for sequential removal of water molecules from gas-phase clusters are summarized in Fig. 1. Although these numbers cannot be

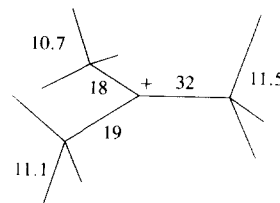


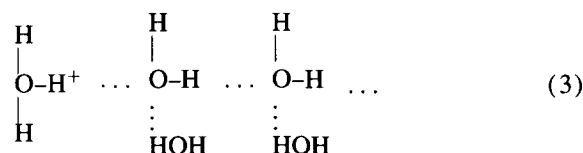
Fig. 1. The structure of H_9O_4^+ , stressing the different coordination states of central and peripheral oxygens. Each vertex represents an oxygen, and each line a hydrogen bond. Hydrogen atoms are not shown. Assuming a cluster with three second shell waters, the numbers (in increasing magnitude) are the sequential bond-dissociation enthalpies (in kcal/mol) from mass-spectrometric data [35].

directly transferred to bulk solution, it is clear that hydrogen bonding to H_3O^+ is considerably stronger than in bulk water. The extra-strong hydrogen bonding correlates with shorter OH...O distances, 2.55 Å compared with 2.8 Å in bulk water. A proton mobility mechanism that involves the rupture of even one of these three bonds will lead to an activation energy exceeding the observed 2–3 kcal/mol value. The first solvation shell of H_3O^+ must remain intact during the proton transfer act [9].

Although the above trends are anything but surprising, they have been largely ignored. Hückel, for example, has suggested the rotation of H_3O^+ as the rate-limiting step [3]. This requires the rupture of all three hydrogen bonds in the first solvation shell. In addition to the huge activation energy, such a rotation is quite unnecessary, as the three hydrogens in H_3O^+ are equivalent (electronic reorganization being much faster than nuclear motion) and any one of them may serve as the mobile proton [5]. The Bernal–Fowler [4] and Conway models [5], assume that a water in the first hydration shell rotates either freely [4] or through an angle of 110° [5]. This requires not only the rupture of one of the strong hydrogen bonds to H_3O^+ , but one or two additional 'ordinary' hydrogen bonds to allow rotation through such large angles. The activation energy for this process may easily exceed 10 kcal/mol.

The proton-relay mechanism is incompatible with solvation requirements for either a gas-phase cluster or a solvated H_9O_4^+ . In the first case the proton cannot transfer from the center to one of the three peripheral water molecules because it will become solvated by one instead of three hydrogen bonds. Ab initio calcu-

lations indeed report [13] that such a process is uphill by over 13 kcal/mol. The proton is 80% localized at the central H_3O^+ unit [13]. When the H_9O_4^+ is solvated, the tetrahedral coordination of the first-shell water ligands (Fig. 1) is unfavorable for proton solvation. Cartoons of hydrogen-bonded networks, which omit the three-dimensional structure as well as some of the hydrogen bonds, may at first look convincing. If, however, one adds the missing hydrogen bonds to the relay scheme of Eq. (1),



it becomes clear that the oxygen sites are inequivalent: all are 4-coordinated except the hydronium which is 3-coordinated, so that no relay is possible.

Zundel and Fritsch [7] stressed the importance of the H_5O_2^+ cation over that of H_9O_4^+ . The proton in-between the two water molecules shows extremely high polarizability, leading to the observation of an IR continuum [7]. This 'Zundel polarization' has been reproduced in ab initio calculations [15]. Driven by fluctuations in the water environment which momentarily makes one or the other oxygen more stable, the structure seems to fluctuate between H_9O_4^+ and H_5O_2^+ [18]. While proton polarization undoubtedly plays an important role in proton mobility, it can only be *part* of the mechanism. First, the triply coordinated H_9O_4^+ is probably the more stable protonated cation [16]. Thus, H_5O_2^+ is an appropriate symmetric structure for the intrinsic transfer step [16], but it can only be an intermediate along the reaction coordinate for prototropic motion. Second, the above picture misses the pivotal role played by hydrogen-bond cleavage in the rate-determining step.

3. Towards an acceptable prototropic mechanism

The above discussion suggests the following criteria for an acceptable molecular mechanism of prototropic mobility:

- Proton mobility is an incoherent proton hopping process.

- The actual proton motion is much faster than solvent reorganization.

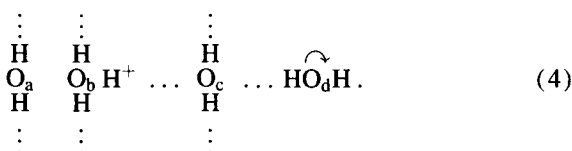
- The rate-limiting step involves cleavage of a single, ordinary hydrogen bond with an activation energy of about 2.6 kcal/mol and a deuterium isotope effect of 1.4.

- The three hydrogen bonds in the first solvation shell of H_3O^+ cannot participate in this step.

- During the elementary transfer step itself, the donor and acceptor water molecules with their solvation shells present equivalent structures.

With these restrictions in mind, it is almost self-evident what the Grotthuss mechanism might be.

Consider a triply coordinated H_3O^+ , i.e. the H_9O_4^+ cation, as the reactant state. Since the rate-limiting step involves cleavage of a hydrogen bond, but not one in the first solvation shell, it is natural to consider hydrogen-bond cleavage in the *second* solvation shell as the rate-limiting step. This leads to isomerization of the H_9O_4^+ cation into H_5O_2^+ , namely



This is depicted in Fig. 2. For clarity, four of the oxygen atoms are identified by corresponding letters.

Initially, the proton is localized on O_b and H_2O_c is one of the three water ligands in the 1st solvation shell, Fig. 2a. Thus $\text{O}_b\text{H} \dots \text{O}_c$ is an extra-strong bond which cannot easily break. H_2O_c has at most one hydrogen bond donating ligand, H_2O_d , which is second-shell with respect to O_b and thus of ordinary strength. The cleavage of the $\text{O}_c \dots \text{HO}_d$ bond is postulated to be the rate-limiting step, Fig. 2b. The reorientation of H_2O_d may be quite small (much smaller than the 110° rotation required by the Conway mechanism [5]), so that just a single hydrogen bond is broken. This may cost 2–3 kcal/mol and take around 1 ps at room temperature.

Following bond cleavage, one expects ultrafast (fs) readjustment of bond angles and bond lengths to form $\text{H}_2\text{O}_b \dots \text{H}^+ \dots \text{O}_c\text{H}_2$. For example, the $\text{O}_b \dots \text{O}_c$ distance should shrink by some 0.15 Å, from 2.55 Å in H_9O_4^+ to the 2.4 Å distance of H_5O_2^+ (these are gas-phase values). At the same time, the distances between

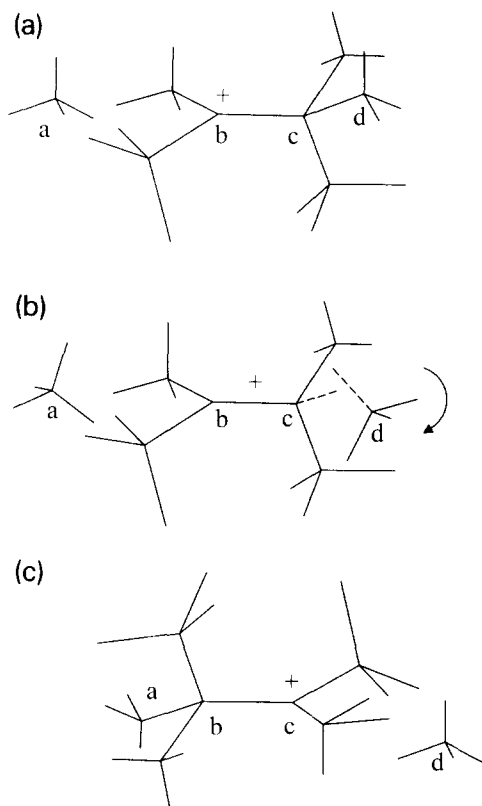


Fig. 2. A molecular mechanism for prototropic mobility. See text for detail.

O_b and the remaining two oxygens are expected to increase from 2.55 Å to below 2.8 Å. Subsequently, fast fluctuations of surrounding water dipoles momentarily stabilize O_b or O_c , leading to large proton polarizabilities within this complex [7,15]. Eventually, one of these fluctuations couples to reorientation of the H_2O_a water, which donates the fourth hydrogen bond to O_b . The proton is now localized on O_c , Fig. 2c. The $H_5O_2^+$ has re-isomerized to a $H_9O_4^+$ cation, centered on a neighboring water molecule. The proton has effectively hopped incoherently across the O–O distance in $H_9O_4^+$.

Proofs of the above mechanism are more likely to come from computational chemistry than from any single experiment. Thus, in their simulation of proton transfer from HCl to water, Ando and Hynes [17] have observed the cleavage of a second-shell hydrogen bond following proton transfer. Tuckerman et al. [18] conducted ab initio molecular dynamics simulations on a 32-water molecule cluster using the Car-

Parrinello density-functional technique. The animated trajectories are qualitatively consistent with the above mechanism. It would be interesting to map-out the complete reaction-coordinate and verify that the dominant barrier is in breaking the second-shell hydrogen bond.

Concluding, proton migration may be envisioned as a process propelled by hydrogen-bond cleavage, taking place in front of the moving proton, and hydrogen-bond formation in its back. If an allegoric description is required, it may be found in Moses crossing the sea, with waves parting before him and reclosing behind his track. It would be amusing if this understanding could be put to practical use. Could proton migration be speeded-up by a moving, hydrogen-bond-cleaving IR wave? The 180 cm^{-1} 'intermolecular' band, thought to involve hydrogen-bond stretch [23], might be a useful frequency to try.

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